

A Robust Parking Detection Algorithm Against Electric Railway Magnetic Field Interference

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Abstract—Wireless magnetic sensor networks (WMSNs) offer an effective and efficient low-cost alternative for real-time parking space monitoring. Current algorithms used for parking space detection based on the WMSNs have been well studied but not without eliminating the geomagnetic field interference. Electric railway systems, such as subways and light rail systems, produce magnetic field radiation that interferes with the geomagnetic field. The electric railway magnetic field interference has increased and is prominent in urban settings; therefore, it is imperative to filter out the interference in order to increase the detection accuracy. This paper proposes an algorithm for parking space detection, which can effectively eliminate the geomagnetic background interference. We solve the parking detection problem through signal filtering technology. Based on mathematical morphology, we designed two filters for extracting morphological structures from interfered magnetic signals. The experimental results verify that the proposed algorithm is more accurate than the existing algorithms.

Index Terms—Vehicle detection, magnetic sensor, interference, morphological filter.

I. INTRODUCTION

CURRENTLY, real-time parking demand and availability has become a major issue in densely populated urban settings since parking spaces are limited and the number of vehicles is continually increasing [1]. Searching for an available parking space during peak hours is usually difficult and leads to additional energy waste, traffic congestion and air pollution. The need for real time parking monitoring systems is significant and continues to be in demand.

Parking detection mechanisms can be categorized into two types: indoor and outdoor. There are various methods for indoor parking detection, for example: video cameras [2], ultrasound [3], and infrared radiation [4]. However, this type of technology is not viable outdoors because installing cables and wires needed for energy supply and data transmission is not cost effective. Thus, a total wireless solution is a feasible option. Wireless Magnetic Sensor Networks (WMSNs) have low power consumption, are small and cost-effective [5].

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Also, they are inherently immune to environmental factors such as rain, wind, snow or fog than traditional wired technologies. Recently an outdoor parking detection WMSN, for about 5,000 parking spaces, is being gradually implemented in Shenzhen, China.

In this parking monitoring system, magnetic sensor nodes are placed at the center of each parking space and the sensor periodically samples the geomagnetic field. When a vehicle enters or leaves the parking space, the parking detection algorithm can detect the change in the parking space status and generates detection events. Detection events data is then collected and transmitted to a server. Having collected the event data from these synchronized sensor nodes, the server can calculate the parking duration of each vehicle. This data can be used for different types of applications, such as parking navigation, parking collection fees, illegal parking detection, etc.

The geomagnetic field is highly susceptible to interference. The operation of electric railway systems, such as subway and light rail systems, produces a wide range of magnetic field radiation, which interferes with the geomagnetic field. Santarelli *et al.* [6] analyzed the geomagnetic background noise at L' Aquila geomagnetic observatory during 2006 and 2007, and they found that the main source of artificial noise is constituted by the Direct Current (DC) electrified railways. Lowes [7] found that DC electric railways produce magnetic fields mainly from unintended earth-leakage currents. Pirjola [8] proposed a computational model for the magnetic field produced by DC railways assuming that the leakage current density changes linearly. Currently, many cities are building subway systems, so the magnetic interference has become a factor to consider when deploying a WMSN.

There are numerous research studies of parking detection algorithms based on wireless magnetic sensor networks. However, these algorithms presume that the background geomagnetic field is without interference. They do not include mechanisms to counter the interference and are prone to produce high false and missed detection rates in environments that have magnetic interference. We propose a parking detection algorithm which can effectively eliminate the geomagnetic field interference. Our main contributions are as follows:

(1) This paper introduces the problems of magnetic field interference caused by electric railway and proposes an “eliminating interference” algorithm for parking detection. To the best of our knowledge, this is the first work to study the geomagnetic interference problem for a WMSN based parking monitoring system.

(2) The parking detection problem is mapped to a signal filtering problem, and morphological filters are designed for extracting target structures from interfered signals.

(3) Experiment results verified that the proposed algorithm can effectively filter and eliminate the magnetic background interference without significantly affecting the lifetime of the WMSN system.

The remainder of this paper is organized as follows. Section II surveys the related works. Section III presents the problem definition. Section IV proposes the algorithm for parking detection. Section V conducts experiments to verify the performance of the algorithm, and Section VI makes a brief conclusion.

II. RELATED WORKS

A considerable amount of research has been conducted on the use of WMSN for traffic information collection. Cheung and Varaiya [9] explored the applications for vehicle detection, speed estimation, and classification. Wang *et al.* [10], [11] proposed a similarity based vehicle detection algorithm to detect vehicles in low SNR (Signal-to-noise ratio) conditions by calculating the correlation [12] between sampled signals and a reference signal. Kwong *et al.* [13] presents a scheme to estimate the travel time distribution of a single arterial link, spanning several signalized intersections. Taghvaeeyan and Rajamani [14] presents a portable roadside magnetic sensor system for vehicle counting, classification, and speed measurement.

Parking detection based on WMSN has been present in both the research community and industry (e.g., ParkSol). Threshold and state machine mechanisms are used to trace the energy change of the magnetic field when a vehicle is parked [15]. If the magnetic field exceeds a threshold, the sensor will indicate the space is occupied. Cheung and Varaiya [9] proposed an adaptive threshold algorithm can be used for parking detection. Ding and Cheung [16] proposed a min-max detection algorithm. Modified min-max algorithm for magnetometers [17] has also been published. Zhu and Yu [18] proposed a parking detection algorithm that combines state machine and cross correlation mechanisms. Zhang *et al.* [19] proposed an algorithm for parking occupancy detection based on the distance discriminate analysis method.

Some works have studied the problem of adjacent interference. Trigona *et al.* [20] conducted the magnetic interference test of the adjacent vehicle in bay parking spaces. Balid *et al.* [21] found that the magneto-resistive sensor node is easily disturbed by the vehicles which have high-intensity magnetic field (such as large-scale trucks, large buses, etc.) on nearby lanes, causing false detection problems and they found that the Z-axis was less affected by interference. Liu and Zheng [22] analyzed the interference problem caused by the neighboring vehicles and studied the influence of parking direction and sensor coordinate direction on the magnetic field strength.

Some researchers propose to add other types of sensors for auxiliary detection. Sifuentes *et al.* [23] have proposed a wireless sensor node based on two sensors (one magnetic and one optical) to detect the presence of a vehicle. When a

vehicle arrives or departs, an optical sensor detects the change in light and wakes up the magnetic sensor which measures the magnetic field in order to detect the vehicle. Huang *et al.* [24] also designed a similar node to solve the adjacent interference problem.

In summary, the vehicle detection for WMSN in the non-interference environment has been studied extensively. Some works have discussed the adjacent interference. However, there is no relevant work to study the issue of geomagnetic field interference caused by an electric railway.

III. PROBLEM DEFINITION

Considering a WMSN $G = (D, R, BS)$, where $D = \{D_1, D_2, \dots, D_n\}$ is the set of sensor nodes, $R = \{R_1, R_2, \dots, R_m\}$ is the set of routers, and BS denotes a base station. Sensor nodes are deployed alongside the roadside and each node is mounted on the center floor of a parking space. For each sensor D_i ($1 \leq i \leq n$), it continuously collects the Earth's magnetic field with sampling frequency f by the magnetic sensor. D_i has running a parking detection algorithm which can be described as $S(k) = IS_Parking(M(k))$, where $M(k)$ is the magnetic field data, and $S(k)$ is a binary result 0 or 1 to indicate the vacant or occupied status of parking space, respectively. If the status is changed, D_i sends an event message $\{D_i, Time(k), S(k)\}$ to its parent R_p , then R_p relays the message to BS through multi-hop.

The measured signal $M(k)$ from a sensor node can be expressed as equation 1,

$$M_i(k) = C_i(k) + I_i(k) + V_i(k) \quad i \in \{x, y, z\} \quad (1)$$

where k denotes the latest sample, $C_i(k)$ is the geomagnetic field which is uniform over a wide area on the scale of kilometers, $I_i(k)$ is the interference magnetic signal, and $V_i(k)$ is the magnetic signal created by a parking vehicle, i indicates the index of axis $\{x, y, z\}$. Usually, the used magnetic sensor is a three-axis AMR (Anisotropic Magneto-Resistive, AMR) sensor [21], [22].

A. Signal Model of Parking

Generally, the magnetic field of a vehicle can be approximated by the field created by a rectangular magnet. Fig. 1 shows a rectangle magnet with length A , width B , height C and a uniform magnetization in its body under the saturation state with symmetrically magnetizing along the Z-axis. As shown in Fig. 1, $P_s = (l, 0, 0)$ represents a sensor node deployed in the parking space, l is the maximum sensing distance. The point $P_v = (0, 0, h)$ is the initial position of a vehicle, where h is the distance from the chassis of the vehicle to the ground.

Given the linear motion of the vehicle along the X-axis, the parking process is as follows: at time $t = 0$, a vehicle starts from P_v and enters the parking space with velocity v_0 , then it parks on the sensor P_s for a period of T , and then it leaves the parking space. Assuming the vehicle's acceleration is uniform during the entering and leaving phases, i.e. a is acceleration and $a = v_0^2/2l$. The trajectory of the vehicle can

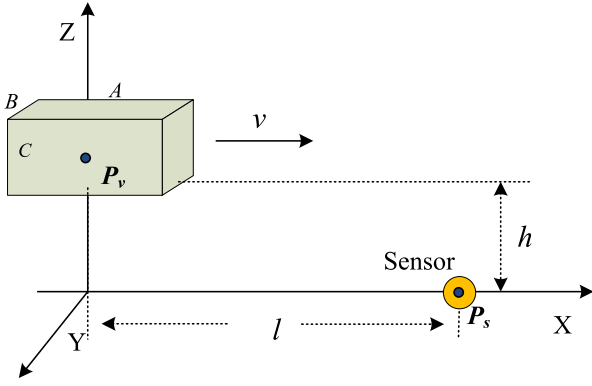


Fig. 1. Model of a vehicle parking.

be expressed as $P_v(t) = \{x(t), y(t) = 0, z(t) = h\}$, $x(t)$ is shown in equation 2,

$$x(t) = \begin{cases} \int_0^t (v_0 - at) dt & \text{if } 0 < t \leq T_e \\ l & \text{if } T_e < t \leq T_s \\ l - \int_{T_s}^t (a * (t - T_s)) dt & \text{if } T_s < t \leq T_l \end{cases} \quad (2)$$

where $T_e = v_0/a$, $T_s = T_e + T$ and $T_l = T_s + T_e$ are the end times of a vehicle's entering, parking and leaving phases, respectively.

According to the magnetic field model of a rectangle magnet [25], the magnetic signal of X-axis created by a vehicle parked on point P_s can be calculated as equation 3,

$$M_x(t) = \frac{K}{2} [\Gamma(l - A/2 - x(t), -B/2, -h) + \Gamma(l - A/2 - x(t), 3B/2, -h) - \Gamma(A/2 - l - x(t), -B/2, -h) - \Gamma(A/2 - l - x(t), 3B/2, -h)] \Big|_0^C \quad (3)$$

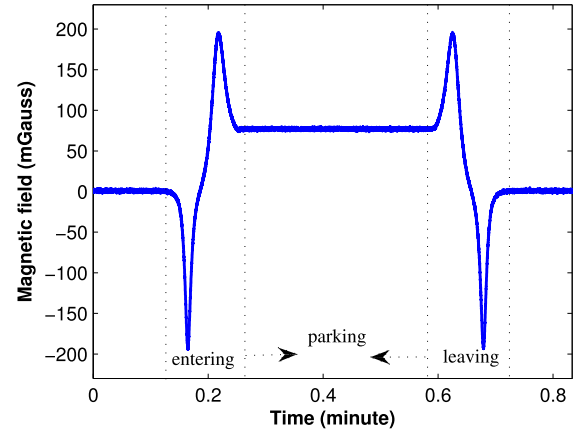
where $K = \mu_0 J / 4\pi$, μ_0 is the air magnetic permeability, J is the surface current density on any plane that is parallel to plane xOy , and

$$\Gamma(r_1, r_2, r_3) = \ln \frac{\sqrt{r_1^2 + r_2^2 + (r_3 - \hat{z})^2} - r_2}{\sqrt{r_1^2 + r_2^2 + (r_3 + \hat{z})^2} + r_2} \quad (4)$$

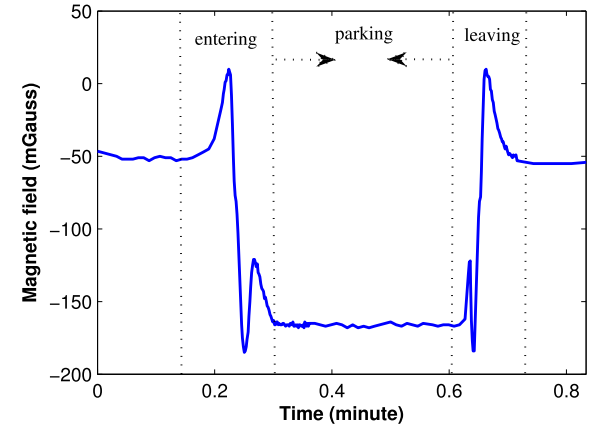
where $[\bullet] \Big|_0^C$ represents the difference of function in the bracket when $\hat{z} = C$ and $\hat{z} = 0$.

Assume the initial speed is $v_0 = 5 \text{ m/s}$, the parking stop time is $T = 20 \text{ s}$, the sampling frequency is $f = 200 \text{ Hz}$, and the measurement noise is Gaussian white noise. Fig. 2(a) shows the simulated signal based on using the magnetic signal model of equation 3. On the other hand, experimental results conducted under similar configuration can be found in Fig. 2(b). Additional examples are provided in Appendix A. The simulated patterns are in agreement with the experimental measurements. We can obtain the signal structure characteristics as shown in definition 1.

Definition 1 (Signal Structure of Parking): As shown in Fig. 2, an entire parking period includes three phases:



(a)



(b)

Fig. 2. Magnetic signals of vehicle parking. (a) Simulation signal. (b) Sensor sampled signal.

entering, parking and leaving. The basic signal structures of each phase are summarized as follows:

(1) *Entering and leaving phases are identified by a set of Gaussian-like pulses. The entering phase is the period when a vehicle enters the sensing range until the vehicle has completely stopped. The signal feature at this phase is positive and negative Gaussian-like pulses [26]. The period from when a vehicle starts the engine until it is beyond the sensing range is the leaving phase. In the leaving phase, the structure of the magnetic signal is the same with the entering phase.*

(2) *Parking phase is identified by a rectangle. The parking phase is the duration of a vehicle, while stopped, in the parking space. In the phase, there is a stable disturbance on the Earth's magnetic field and the structure of the signal is like a rectangle.*

B. Magnetic Interference

The electric railway's circuit, in its basic structure, consists of a rectifier substation, an overhead power line, a train, the rails and the ground to conduct the return current, as shown in Fig. 3. The magnetic fields [27] produced by the system mainly include three types: First, the trains themselves produce

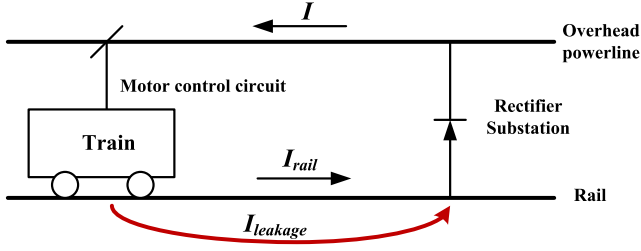


Fig. 3. The circuit of an electrified subway.

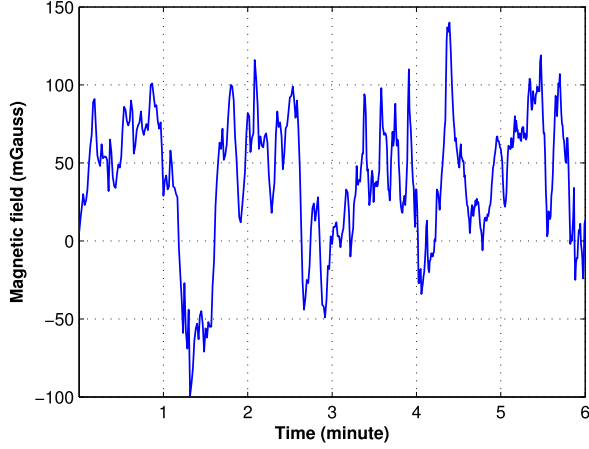


Fig. 4. The interference signal created by the subway.

local magnetic fields. Second, the magnetic field produced by the traction current in the loop substation-overhead-train-rails-substation will produce a magnetic field. Third, the magnetic field due to currents that have leaked from the track into the ground is significant. The running rails are close to the ground, so unless precautions are taken, it is very easy for them to have significant electrical contact with the ground. The first and second types of magnetic field are significant only when observing on or near the train. But currents leaking unto the ground from the systems produce a direct effect over tens of kilometers on the Earth's magnetic field [28].

We deployed sensor nodes on the roadside parking spaces nearby a subway. For example, Fig. 4 shows the interference signal collected by the x-axis $I_x(k)$. $I_x(k)$ is a random fluctuation signal and the magnitude of fluctuation greater than the normal geomagnetic changes which is less than 5 mGauss. The probability density function of $I_x(k)$ can be approximated by the method of fitting with the normal distribution curve, as shown in equation 5,

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (5)$$

where μ is the mean or expectation of the distribution, σ^2 is the variance. Using the sample data as shown in Fig. 4, we can calculate $\mu = 25.0$ and $\sigma^2 = 1301.0$, and the theoretical distribution is shown in Fig. 5.

Fig. 6 shows the probability density distribution of the magnetic signal created by parking vehicles. The magnetic field is distributed between -250 mGauss and 200 mGauss. Comparing with Fig. 4, we can see that the interference signal

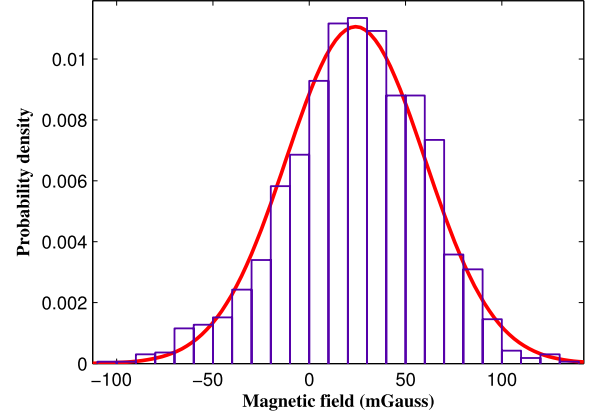


Fig. 5. The probability density of the interference signal.

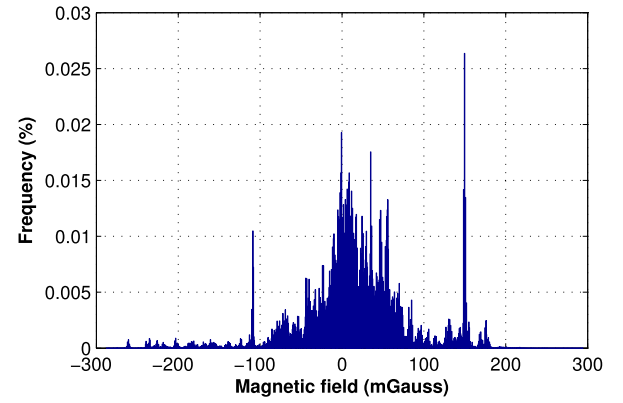


Fig. 6. The probability density distribution of the target magnetic signal.

is significant; therefore it is imperative to filter out noise in order to increase the detection accuracy.

According to the above description, the parking detection problem can be defined as:

Definition 2: Parking Detection:

$$S(k) = IS_Parking(M(k))$$

Input: $M_i(k) = C_i(k) + I_i(k) + V_i(k) \quad i \in \{x, y, z\}$

(1) $C_i(k)$ is the Earth's magnetic field.

(2) $V_i(k)$ is the target magnetic signal of a vehicle. A whole parking process can be divided into three phases whose structure characteristics are defined in the definition 1.

(3) $I_i(k)$ is the interference magnetic signal created by electric railway system.

Task: The algorithm needs to accurately detect the structural characteristics of the three parking phases from the interfered signals. And according to these characteristics, it can make an estimation of the status of the parking space.

Output:

(1) $S(k) = 1$: if the Gaussian-like pulse and rectangular structures are both detected, the algorithm output is $S(k) = 1$, it describes the status of the parking space is {occupied}.

(2) $S(k) = 0$: when both Gaussian-like pulse and rectangular structures are disappearing, the algorithm output is $S(k) = 0$, that is, the parking space is {vacant}.

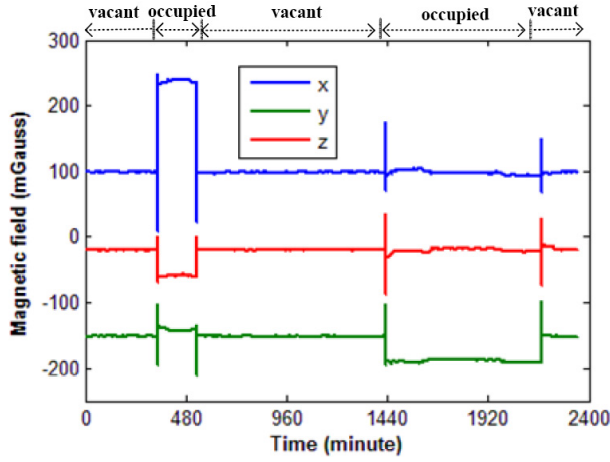


Fig. 7. The magnetic signals of parking events.

The main task of the algorithm is to accurately extract two types of structural signals from the interfered signal: Gaussian-like pulse and rectangular structure. We adopt a mathematical morphology theory [29] to design signal filters which can effectively filter noise and extract morphological structures. The advantage of mathematical morphological filtering is that the algorithm is simple and requires less computation. It does not need to consider the type of noise and only needs to select structural elements that match the target signal. This method is suitable for sensor nodes to process signals in real time.

IV. ALGORITHM

In this section, we first introduce the signal synthesis, then explain how to design morphology filters for Gaussian-like pulse and rectangular structure detection, and then describe the pseudo code of the parking detection algorithm.

A. Signal Synthesis

A 3-axis AMR (Anisotropic Magneto-Resistive) magnetic sensor has been used. Detection with single axis or dual axis data may result in missed or false detections. As shown in Fig. 7, a node has sensed two parking events. In the first parking event, the change of y-axis signal is much less than that of x- and z-axis so that detection with y-axis will result in a missed detection. However, in the second parking event, only the y-axis varied about 45 mGauss. If x- and z-axis are used for detection, it may result in a missed detection.

It is necessary to integrate three axis data for detection. In order to reduce the computational complexity and reflect the variation of the three-axis, the three-axis signals are synthesized as a fused signal using equation (6),

$$f(k) = \sum_i W_i(k) * |M_i(k) - C_i(k)| \quad i \in \{x, y, z\} \quad (6)$$

where W_i is the weight value of i -axis, C_i is the Earth's magnetic field.

The interference of the geomagnetic field is dynamic. For example, Fig. 8 shows the interference signal in one day collected by a sensor deployed on a road near the subway.

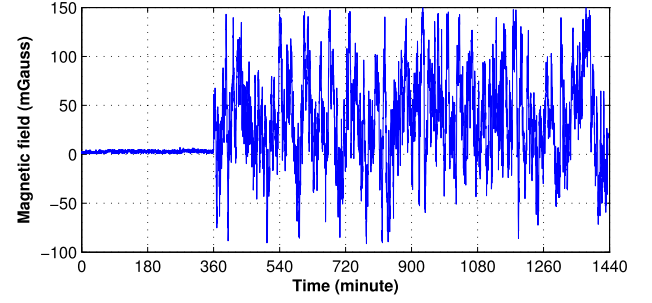


Fig. 8. The interference signals of a node in one day.

The service time of the subway is from 6:00 to 24:00 and the out of service time is from 0:00 to 6:00. The magnetic interference is significant during the service time and there is no interference during the out of service time.

The calculation of W_i includes two cases: the first case, if there is geomagnetic interference, W_i is related to the degree of the interference, the smaller the interference, the greater the proportion; the second case, if there is no geomagnetic interference, $W_i = 1$,

$$W_i(k) = \begin{cases} (Sum(k) - \lambda_i(k)) * 1.5 / Sum(k) & \text{if } \exists \lambda_i(k) > 5 \\ & i \in \{x, y, z\} \\ 1 & \text{otherwise} \end{cases} \quad (7)$$

where $\lambda_i(k)$ is the sample variance value, $Sum(k) = \sum_i \lambda_i(k)$;

$$\lambda_i(k) = \begin{cases} \frac{1}{N-1} \sum_{j=k-N}^{k-1} (M_i(j) - \mu_i(k))^2 & \text{if } S(j) = 0 \\ \lambda_i(k-1) & \text{Otherwise} \end{cases} \quad \forall j \in [k-N, \dots, k-1] \quad (8)$$

where μ is the sample mean and N is the sample size. The sample variance is updated by the sensor reading when there is no vehicle is detected within the duration of N samples.

Earth's magnetic field has been drifting westward [30]. For a long-running parking detection system, for example 5 years, the change of the Earth's magnetic field cannot be ignored. Conversely, Earth's magnetic field is uniform over a wide area on a scale of kilometers [9]. The data we collected in our experiments showed that the Earth's magnetic field was almost constant over a period of one month. In our system, BS instructs the nodes to update C_i on a monthly basis. C_i is approximately equal to the average of the samplings when the geomagnetic field has not interfered and the status of the parking spaces is *vacant*.

B. Gaussian-Like Pulse Filter

The Top-hat transformation in mathematical morphology can be used to detect the crest or trough of the signal to be processed. It has been widely used in the fields of infrared target detection, license plate location, image analysis and

pattern recognition [31]. The filter based on Top-hat operation to extract the Gaussian-like pulses is shown in equation (9);

$$TH(k) = OHat(f(k), g(m)) + CHat(f(k), g(m)) \quad (9)$$

where $f(k)$ is the input signal, $g(m)$ is a structure element, $OHat$ is the open Top-hat transform, and $CHat$ is the close Top-hat transform.

Commonly used structural elements are rectangular, linear, round, triangular and so on. Previous studies [29] have shown that it is necessary to select structural elements matching with the target signal's characteristics. Filters need to remove interference and extract fluctuation signal which has Gaussian-like characteristics. So we chose a Gaussian function (10) to construct the structural element,

$$g(m) = \frac{P}{\sqrt{2\pi}\sigma^2} e^{-\frac{(m-L/2)^2}{2\sigma^2}} \quad 0 \leq m \leq L \quad (10)$$

where P is a given amplification factor, σ^2 is the variance, L is the length of the structural element and it is closely related to the length of the Gaussian structure in the signal to be extracted.

According to the different form of open and close operations, the Top-hat transform can be divided into two types: open Top-hat transform and closed Top-hat transform [32], and they are defined as equations (11) and (12), respectively;

$$OHat(f(k), g(m)) = f(k) - (f(k) \circ g(m)) \quad (11)$$

$$CHat(f(k), g(m)) = f(k) - (f(k) \bullet g(m)) \quad (12)$$

where $\circ$ and $\bullet$ are the basic opening and closing operations, respectively. For the basic operations of mathematical morphology, please refer to Appendix B.

Since g is a Gaussian structure, the operation $OHat$ can effectively extract the positive Gaussian-like pulse from f , and $CHat$ is used to extract negative Gaussian-like pulses. So $TH(k)$ is the extracted positive and negative Gaussian-like pulses from signal $f(k)$. The function $y(k) = \text{Binarize}(TH(k), T_1)$ binarize the signal $TH(k)$, where T_1 is a threshold value, and $y(k)$ is the detection result of the Gaussian-like pulses. For example, Fig. 9 is the initial input signal $f(k)$. The extracted Gaussian-like signal $TH(k)$ from $f(k)$ is shown in Fig. 10. Fig. 11 are the detection results $y(k)$ from the initial signal $f(k)$. $y(k)$ is a time series of 0 and 1. To facilitate viewing the detection results in the graph, we enlarge 1:180.

C. Rectangular Structure Filter

The filter to extract the rectangular structure is shown in equation (13), where $f(k)$ is the input signal, $h(m)$ is a structure element. The morphological operation OC (or CO) is the combination with opening (or closing) operator followed by closing (or opening) operator, which is used to eliminate pulse interference and extract the rectangular structure signals.

$$MA(k) = \frac{OC(f(k), h(m)) + CO(f(k), h(m))}{2} \quad (13)$$

The selection of structural elements is closely related to the waveform of the signal. The signal in the parking phase

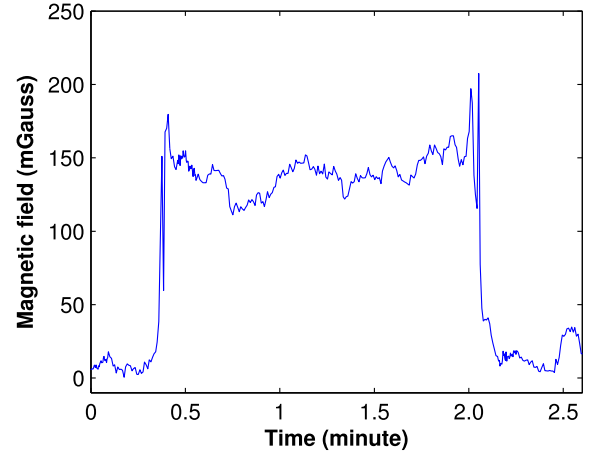


Fig. 9. The initial input signal.

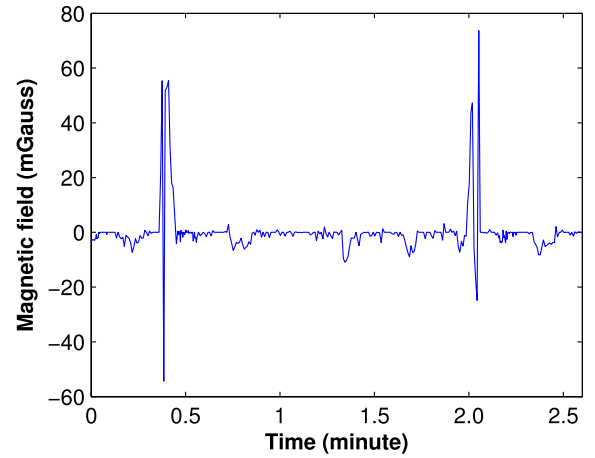


Fig. 10. The extracted Gaussian signal.

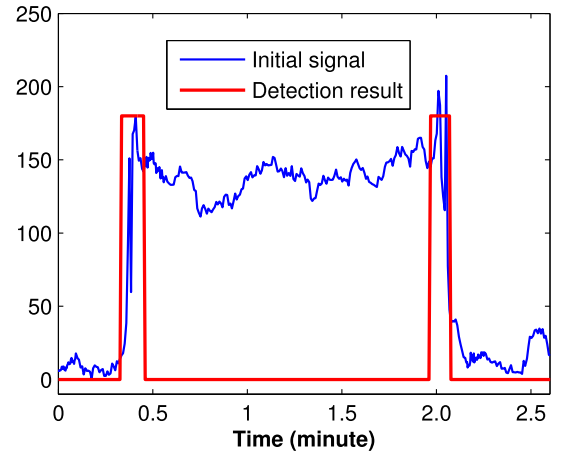


Fig. 11. The detection result.

is similar to the rectangle. So using a rectangle as structure element $h(m)$ (14) to eliminate the interference,

$$h(m) = H \quad 1 \leq m \leq W \quad (14)$$

where H is a constant, W is the length of structural elements, $W_{min} \leq W \leq W_{max}$. The magnetic signals created by electric

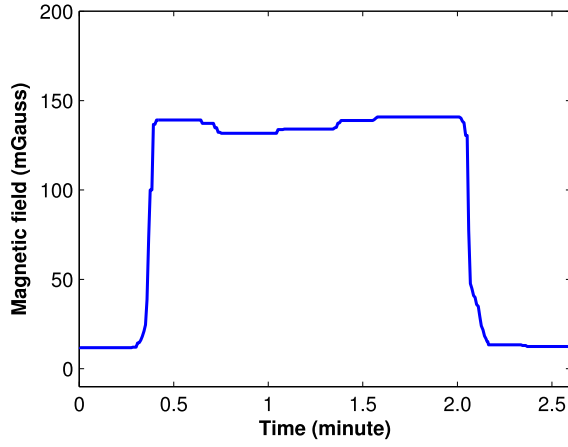


Fig. 12. The extracted rectangle structure signal.

railway contain positive and negative triangle pulse. The width of most of the pulse is between $10 \sim 20$. In order to remove interference, W must greater than the width of pulses, so $W_{min} \geq 20$. There is also a limit to the upper of W . Assume that the parking time of the vehicle is $T_{parking}$, then the length of the rectangle signal to be extracted is $W_{parking} = T_{parking} * f$, f is the sampling frequency. If $W > W_{parking}$, the target rectangle signal will be filtered out as noise, so $W_{max} < Minimum(W_{parking})$. Usually, the height of the structural element H has no effect on the performance of the filter, so a constant value is taken.

Using the structural elements of the same type and size, the morphological opening and closing operations are cascaded and combined in different orders to construct the classic Open-closing and Close-opening filters [29]. They are defined as follows:

Open-closing operation:

$$OC(f(k), h(m)) = f(k) \circ h(m) \bullet h(m) \quad (15)$$

Close-opening operation:

$$CO(f(k), h(m)) = f(k) \bullet h(m) \circ h(m) \quad (16)$$

From the above definition, both OC and CO filters can filter positive and negative pulse noise in the target signal. However, they all lead to statistical deviations in amplitude, i.e., the result of open-closing operation has the lower amplitude than the original signal and the result of close-opening operation has the larger amplitude than the original signal. This may affect the result of rectangle structure detection. So a mean weighted combination method is adopted as equation (13).

For example, Fig. 9 is the initial input signal $f(k)$ which is created by a vehicle parked. The extracted rectangle-like signals $MA(k)$ from $f(k)$ is shown in Fig. 12. Then used the function $r(k) = Binarize(MA(k), T_2)$ to binarize the signal $MA(k)$, where T_2 is a threshold value, and $r(k)$ is the detection result of the rectangle signal. Fig. 13 are the detection results $r(k)$ from the initial signal $f(k)$. $r(k)$ is a time series of 0 and 1. To facilitate viewing the detection results in the graph were enlarge 1:150.

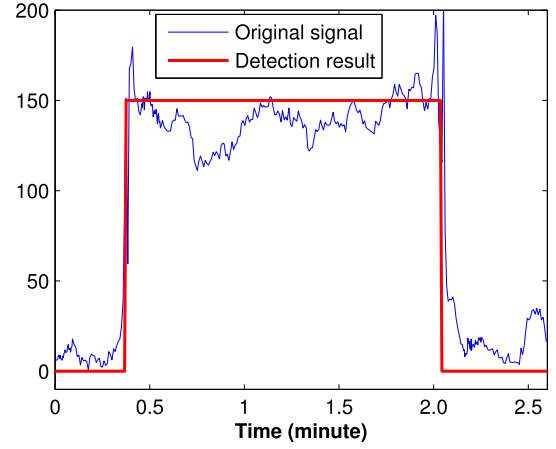


Fig. 13. The detection result of the rectangle structure signal.

D. Pseudo Code of Algorithm

According to the definition of parking detection problem 2, the pseudo code of the AMID (Against Magnetic Interference Detection) algorithm is described as shown in algorithm 1.

Algorithm 1 AMID Algorithm

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1 Function  $S(k) = IS\_Parking(M_i(k))$ 
   Input : The original samples  $M_i(k)$   $i \in \{x, y, z\}$ 
   Output: The status of parking space
            $S(k) = 0$  is vacant
            $S(k) = 1$  is occupied
   /* Signal synthesis */
2  $f(k) \leftarrow \{M_x(k), M_y(k), M_z(k)\};$ 
   /* Gaussian-like pulse filter */
3  $g(m) \leftarrow a \text{ Gaussian structure element};$ 
4  $TH(k) = OHat(f(k), g(m)) + CHat(f(k), g(m));$ 
5  $y(k) = Binarize(TH(k), T_1);$ 
   /* Rectangle structure filter */
6  $h(m) \leftarrow a \text{ rectangle structure element};$ 
7  $MA(k) = (OC(f(k), h(m)) + CO(f(k), h(m)))/2;$ 
8  $r(k) = Binarize(MA(k), T_2);$ 
9 if  $y(k) == 1$  and  $r(k) == 1$  then
10 |  $S(k) = 1;$ 
11 else if  $S(k-1) == 1$  and  $r(k) == 1$  then
12 |  $S(k) = 1;$ 
13 else if  $y(k) == 0$  and  $r(k) == 0$  then
14 |  $S(k) = 0;$ 
15 else
16 |  $S(k) = 0;$ 
17 end

```

V. EXPERIMENTS

In this section, we evaluate the performance of AMID through experiments. First, we introduce the experimental scenarios and results of a deployed WMSN. Then the performance comparison with the existing parking detection algorithms is carried out.

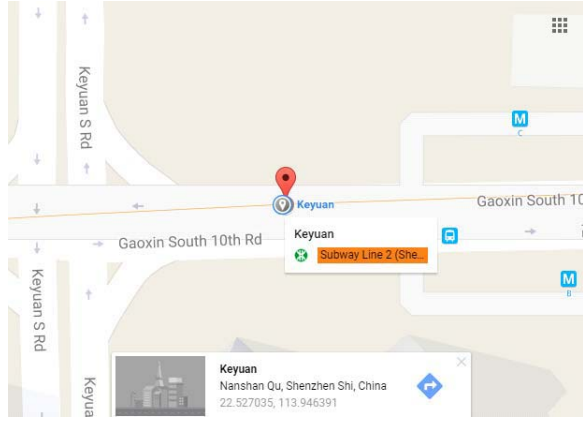


Fig. 14. The location of experiments.

A. Experimental Scenario and Results

As shown in Fig. 14, the selected roadside parking areas are in the Gaoxin South 10th Rd, Shenzhen, China, which is above the subway Line 2. The subway line 2 is a 60 km rapid transit system and its highest speed is 80 km/h. Trains are fed at 1500 V direct current through overhead wires. The subway line's train departure interval is between 3-6 minutes.

To test the proposed algorithm, we deploy sensor nodes in three types of parking: parallel, bay and angled. All of these parking areas are near the subway line 2. As shown in Fig. 15(a), the node uses the STM8L151C8 MCU, has LoRa SX1278 [33] communication module and QMC5883L [34] magnetic sensor. The node runs the Contiki operating system [35] and its shell uses a high-strength PC material for resisting the weight of vehicles. The routers and BS use wireless communication and are solar powered. As shown in Fig. 15(b), each of router and BS is equipped with a 44 Ah lithium battery pack and a solar panel. The routers and BS are attached on roadside lampposts to improve the quality of wireless communication and solar charging efficiency.

In our experiments, parameters related to the parking detection algorithm are given in table I. We analyzed the performance of the AMID using the accuracy of detection as in definition 3.

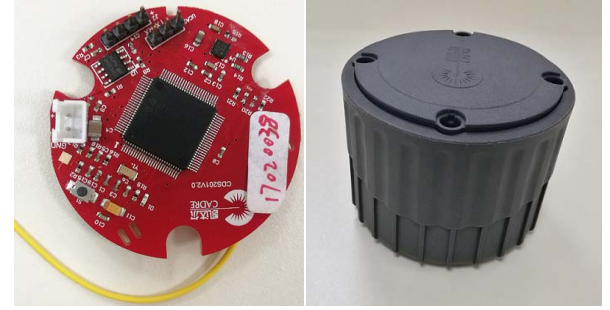
Definition 3 (Accuracy of Detection): The lifetime of sensor node D_i can be divided into an event stream according to the status of its corresponding parking space,

$$Life(D_i) = \{t_i(v_1), t_i(o_1), t_i(v_2), t_i(o_2), \dots, t_i(v_j), t_i(o_j)\} \quad (17)$$

where $t_i(v_j)$ is the duration of the vacant event v_j , and $t_i(o_j)$ is the duration of the occupied event o_j . It is an incorrect detection that the number of missed or false detection in the period of the event v_j or o_j is equal or greater than 1. The correct detection rate of a WMSN is defined as equation (18),

$$Accuracy_Rate = \frac{\sum_{i=1}^N (1 - \frac{Incorrect_i}{Event_i})}{N} \quad (18)$$

where $Incorrect_i$ is the number of incorrect detections of node D_i , $Event_i$ is the number of vacant and occupied events of node D_i



(a)



(b)

Fig. 15. The sensor node and BS. (a) Sensor node. (b) BS.

TABLE I
EXPERIMENT PARAMETERS

Parameter	Description	Value
P	The amplification factor	500
N	The sample size for calculate sample variance	30
L	The length of Gaussian structural element	15
W	The length of rectangular structural element	30
σ	A parameter of Gaussian structure	4.3
T_1	Binary threshold about Gaussian signal	45
T_2	Binary threshold about rectangle signal	50
H	The height of the rectangular structure element	5
M	The length of the buffer G_{buf}	30

In order to verify the proposed AMID algorithm, a wireless magnetic sensor network was installed and operated for about two months. Using manual recording and video monitoring, we obtained the exact status of the parking spaces. Experiment results show that the average detection accuracy of the WMSN in the parallel parking spaces is 98.5%, and in the bay or angled parking spaces is 96.6%. In addition to subway interference there is also interference signals from adjacent vehicles. For example, Fig. 16 shows magnetic signals with adjacent parking space interference. The lower accuracy of the bay and angled parking spaces is mainly due to interference which is produced by a vehicle parked in the adjacent space.

Interference from nearby traveling vehicles produces a short period of fluctuations. Fig. 17 shows the magnetic interference generated by a bus and a truck as they passed through an adjacent lane. The characteristic of a rectangular structure

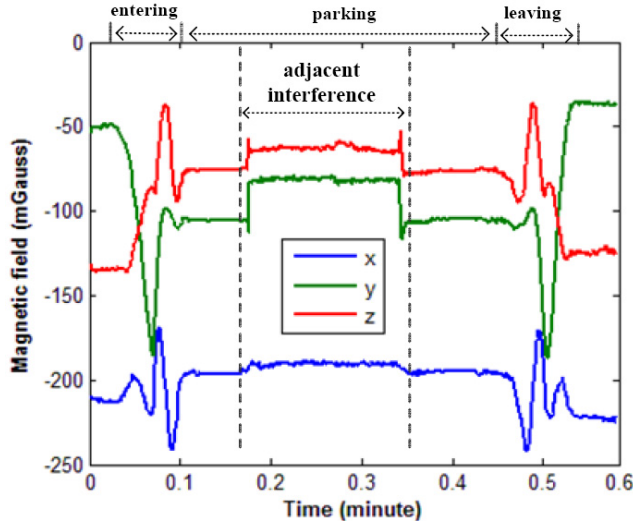


Fig. 16. The magnetic signals with adjacent interference.

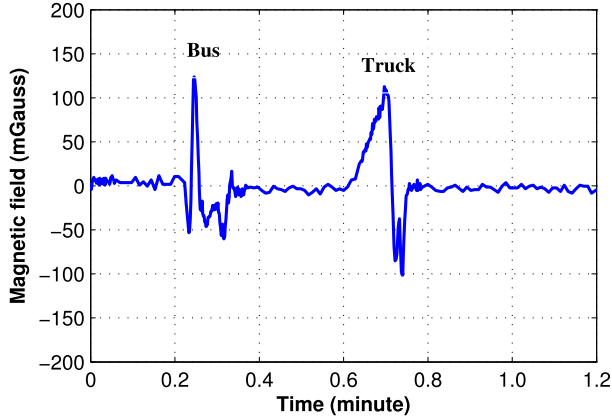


Fig. 17. The interference magnetic signal created by nearby travel vehicles.

is a necessary condition for parking detection, so that the interference of traveling vehicles will not affect the parking detection.

Since the sensor node is battery powered, the battery life span is low, so the majority of the time the sensor node should be in a sleep mode. The power consumption of the node includes sampling, calculation, sending, receiving and sleeping. We can use the time complexity of the parking detection algorithm to evaluate the energy consumption of calculation. The MCU controls the AMR sensor to sample by a pre-setting sampling period. After each sampling, the MCU needs to call the AMID algorithm for calculation. AMID mainly includes two filtering modules: Gaussian filter and rectangular filter. For the Gaussian filter module, the structural element used is a Gaussian signal, and the time complexity of the Open and Close operations is $O(L^2)$, where L is the length of the structure element; For the rectangular filter module, each value of the structural element is equal, which simplifies the calculation, the time complexity of Open and Close is $O(1)$. So the time complexity of AMID is $O(L^2)$.

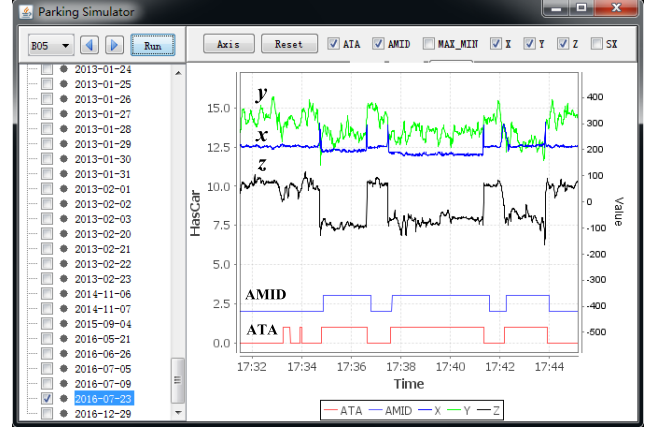


Fig. 18. An algorithm verification system.

In our experiment, the length of the Gaussian structure element is $L = 15$. So the energy consumption of the AMID algorithm is minimal. In the designed WMSN, the node is equipped with a pair of 3.6 V 8000 mAh lithium batteries and it can continuously work for 5 years without changing the battery.

B. Performance Comparison

In order to facilitate the performance comparison of multiple algorithms, we implemented an algorithm verification system. The original sampling data of the magnetic field is packaged and uploaded. Every 30 samples are packed into a packet. The packaged method can reduce transmission conflict and energy consumption. The system receives the magnetic field data uploaded by the BS, realizes data parsing and stores the data in the database. We use C language to implement multiple algorithms. Using the original magnetic field signal obtained from the experiment, algorithms are running on the server side. The interface of the system is shown in Fig. 18, the system can select the magnetic signals according to the node number and date. It can choose to display original magnetic signals and the detection results of the running algorithms.

The threshold and state machine mechanisms based algorithms designed for parking detection such as ATA (Adaptive Threshold Algorithm) [9], MAX_MIN [17] are considered. However, they are not suitable to direct calculation of interfered magnetic signals. To evaluate these algorithms we need to preprocess the signal. First, the three-axis signals are synthesized as a fused signal $f(k)$ by equation (6). Then a smoothing filter as equation (19) takes a running average of the signal $f(k)$,

$$A(k) = \begin{cases} \frac{f(k) + f(k-1) + \dots + f(1)}{k} & \text{for } k < M \\ \frac{f(k) + f(k-1) + \dots + f(k-M+1)}{M} & \text{for } k \geq M \end{cases} \quad (19)$$

where M is the pre-defined length of the running average. For implementation, using a buffer G_{buf} stores the values of $[f(k), f(k-1), \dots, f(k-M+1)]$. $A(k)$ is used as input to simulate ATA and MAX_MIN algorithms.

TABLE II
PERFORMANCE COMPARISON OF ALGORITHMS

Type of parking lots	Algorithm	Detection accuracy
Bay	ATA	75.9%
	Max_Min	84.0%
	AMID	97.5%
Parallel	ATA	79.5%
	Max_Min	85.9%
	AMID	98.5%

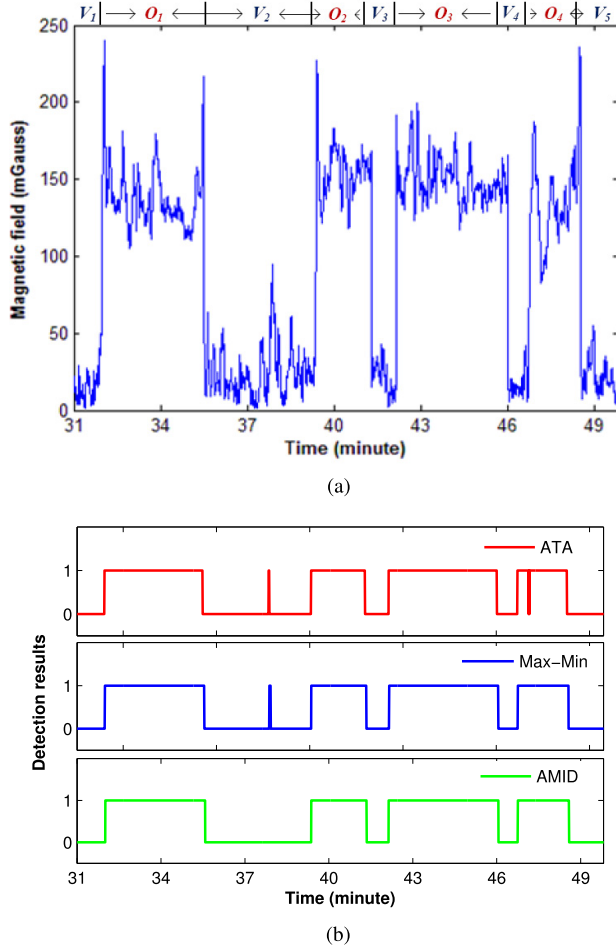


Fig. 19. Parking detection examples of three algorithms. (a) Initial signal. (b) Detection result.

We performed performance comparison tests on different types of algorithms. For each experiment, the sample data we selected contained 1000 parking events. As shown in table II, in both types of parking lots, the detection accuracy of AMID algorithm proposed in this paper is better than ATA and MAX_MIN. Compared with the parallel parking, the detection accuracy of these algorithms has slightly decreased. This is due to the fact that the magnetic signals in the bay parking maybe disturbed by adjacent interference.

Fig. 19 shows an example of parking detection. The initial magnetic signal is shown in Fig. 19(a). The actual event status has been identified above the figure, v_i is the vacant event, and

o_i is the occupancy event. So the initial signal represents a list of events $E = \{v_1, o_1, v_2, o_2, v_3, o_3, v_4, o_4, v_5\}$. The detection results of three algorithms are shown in Fig. 19(b). There are fault detections of event v_2 for both ATA and MAX_MIN algorithms. In addition, there is a missed detection of event o_4 for ATA algorithm. In this example, the AMID algorithm has all been correctly detected.

VI. CONCLUSION AND DISCUSSION

Wireless magnetic sensor networks are being gradually used for outdoor parking detection. However, the geomagnetic field is highly susceptible to interference from electric railway systems. This work proposes a parking detection algorithm which can filter and eliminate the geomagnetic interference. Two filters based on mathematical morphology are proposed to remove the magnetic interference and extract the structural signals of vehicle parking detection. Experiment results show that the detection accuracy of the algorithm is more accurate than previous algorithms for parking detection.

For the future study, we plan to measure another magnitude which is not affected by magnetic interference to improve the detection accuracy.

(1) We can add a second sensor for auxiliary detection, for example, a Light Dependent Resistor (LDR) to measure optical changes of light, or sensors such as infrared, ultrasonic, Doppler radar, and so on. However, these sensors are often susceptible in outdoor environment, such as weather conditions, occlusions, and lighting variations. Conversely, if a second sensor is used, the design of the node's shell needs to be changed. For example, if an optical sensor is added, the diffusion of light should be considered in the design of the shell. We need to conduct further research on the feasibility and cost effectiveness of adding a second sensor.

(2) The change of RSSI (Received Signal Strength Indication) of the node's wireless communication can also be used as a reference for detection. In the process of a vehicle entering and leaving the parking space, if the sensor node tries to send and receive data packets multiple times, its parent node and itself will both sense that the RSSI has changed significantly. However, passing vehicles, pedestrians, and adjacent vehicles' parking will also change the value of RSSI. Therefore, it is necessary to further explore the relationship between the change of RSSI and parking detection.

APPENDIX A MAGNETIC SIGNALS

Magnetic signals are affected by various factors such as the type of parking spaces, the trajectory of the vehicle, and the speed of travel. For example, Fig. 20 shows the magnetic signals of vehicles parked in a bay, an angled and a parallel parking spaces, respectively. Although these signals are different, they share common characteristics as described in definition 1.

APPENDIX B BASIC MORPHOLOGICAL OPERATORS

Mathematical morphology is a theory for the analysis and processing of geometrical structures. Let f and g denote

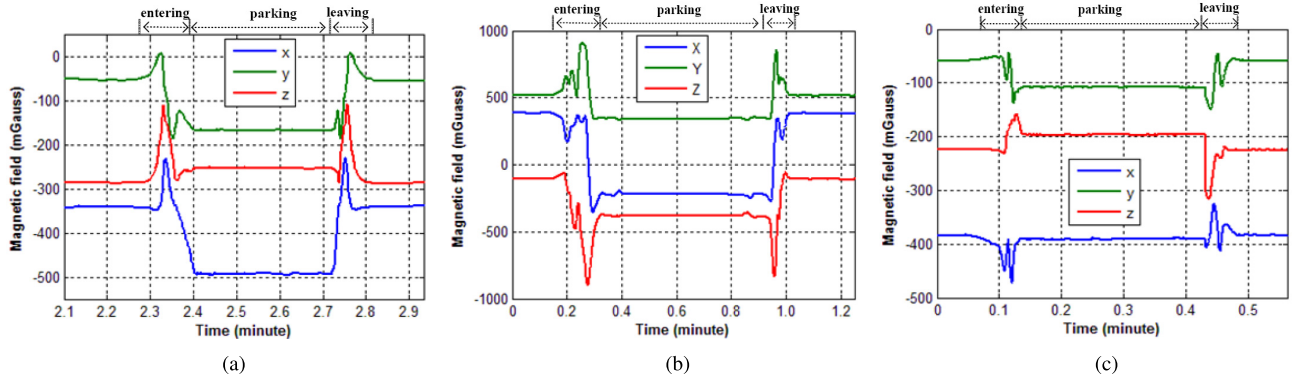


Fig. 20. The magnetic signals sampled from different kinds of parking spaces. (a) Bay parking. (b) Angled parking. (c) Parallel parking.

two discrete-valued functions defined on a one-dimensional discrete space. f is the input signal and g is a structure element. There are four basic morphological operators as follows:

A. Erosion

The erosion of f by a structure element g is

$$(f \ominus g)(k) = \min_{m=1,2,\dots,M} \{f(k+m) - g(m)\} \quad (20)$$

where $k \in (1, 2, \dots, N - M)$.

B. Dilation

The dilation of f by a structure element g is

$$(f \oplus g)(k) = \max_{m=1,2,\dots,M} \{f(k-m) + g(m)\} \quad (21)$$

where $k \in (M, M + 1, \dots, N)$.

C. Opening

Opening of signal f by a structure element g consists of an erosion followed by a dilation. That is,

$$(f \circ g)(k) = [(f \ominus g) \oplus g](k) \quad (22)$$

D. Closing

Closing is a dual operation of opening and consists of a dilation followed by an erosion. The closing of a signal f by a structure element g is

$$(f \bullet g)(k) = [(f \oplus g) \ominus g](k) \quad (23)$$

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